Water Resources and Pollution in Canada

Although water covers 75% of the earth's surface, much of it is in oceans and seas, which contain high mineral salt content. Humans and many other life forms cannot use this salt water to hydrate their living cells. It is therefore essential for us to have fresh water, that is, water with very low mineral salt concentrations. Of all the water on the earth, only 2.7% is fresh water thus making it a precious commodity. Of that, Canada has almost 10% of the world supply. This water is held in the Great Lakes and the numerous inland lakes and rivers that constitute our landscape.

Within Canada, the St. Lawrence River, which drains the Great Lakes, has the largest discharge of fresh water and the Mackenzie River in British Columbia is the longest. Many of Canada's largest lakes are found on the edge of the Canadian Shield. These lakes were gouged into the landscape as the glaciers moved off the bedrock and pushed into the sedimentary rock surrounding the Shield. Great Bear Lake, Great Slave Lake, Lake Winnipeg and four of the five Great Lakes (Huron, Superior, Ontario and Erie) are found on the edge of the Canadian Shield.

Amazing facts and figures about fresh water in Canada.

- In Canada there are 755,180 km^2 of freshwater lakes, ponds and rivers (>100 m. wide).
- The Mackenzie River has the largest drainage area at 1, 805, 200 km$^{2.}$
- The Mackenzie River is the longest river in Canada at 4241 km.
- The St. Lawrence River has the greatest annual discharge at 9 850 m^3
- Wollaston Lake, Saskatchewan, at 2681 km^2, is the largest lake in the world that drains in two directions, north to the Mackenzie River basin and east to Hudson Bay.

Not all-fresh water comes from rain. Glaciers and ice fields produce the largest portion of fresh water, followed by underground aquifers, lakes and rivers and lastly, precipitation. Most fresh water that comes from meltwater or precipitation is collected in the major watersheds. Acting as large catchments, water drains off the highland surfaces into a network of increasingly larger streams and rivers, which eventually empty into the oceans. Canada has four major *drainage basins* or watersheds; three that flow directly into oceans, one that flows into Hudson Bay which eventually also empties into the Atlantic Ocean. One smaller, yet very important watershed within Canada flows through the United States into the Gulf of Mexico from central Canada.

The five major watersheds of Canada are fed by major Canadian rivers.

Yukon River
Columbia River — Pacific watershed
Fraser River

Mackenzie River — Arctic watershed

Nelson River
South Saskatchewan River — Hudson Bay watershed

St. Lawrence River — Atlantic watershed

Portions of
Milk River
Frenchman River — Gulf of Mexico watershed

Water that is caught in depressions and lowlands produces swamps, bogs and marshes known as *wetlands*. These, as well as other large flat areas, which do not drain off quickly, serve as recharge areas for groundwater. Because this water cannot run off into lakes and rivers, it is strained through the surficial material into underground aquifers. These aquifers extend from the bedrock upward. The level to which the water rises from the bedrock in the ground is called a *water table*. This water source is reached by means of wells drilled into the ground and the water is pumped to the surface. Wells are the only source of water for many areas that are at a distance from rivers or lakes.

Because water is such an important resource, man has endeavored to control it. Not only can it be used for human and animal consumption, but also for irrigation, mining, manufacturing and electrical power generation. Man has constructed reservoirs and dams to contain the water. Benefits are numerous in the form of a constant water supply, electricity production, control of flood conditions and recreational uses. However, the construction of a dam or reservoir is destructive as well. Land up river is flooded with respective damage done to vegetation, forests and the need to relocate humans and animals. Changes in water levels and in water flows destroy natural habit for many fish as well as impacting their spawning grounds. In Canada, every major river has some form of water control in effect.

As a study example, let's look at the Ottawa River drainage system. The Ottawa River is 1271 km. long. It collects water from many lakes and streams in its *headwaters* in the upper Ottawa valley and has a total drainage area of 146 300 km^2. At Ottawa it is fed by the Rideau River (drainage area of 3800 km^2) and the Gatineau River (drainage area of 23 700 km^2). It is later joined by the Rivière du Lièvre and the South Nation River (drainage area of 3830 km^2). The Ottawa River empties into the St. Lawrence River at the Lake of Two Mountains just west of the island of Montréal.

Pollution

Water in Canada is often taken for granted and the necessary precautions to protect it are not always followed. Conflicts between man and nature are inevitable. Most of Canada's population lives on the edge of major rivers or lakeshores. Our water supply is often drawn from these same lakes and rivers. Those who live away from rivers, such as sections of the prairie region, draw their water from aquifers. However, we cannot take clean, potable water for granted. Where there is human habitation there is often accompanying waste products in the form of garbage, sewage, industrial discharge and smog. This contamination of our clean environment is called *pollution*. Some pollution is very visible such as a waste disposal site. But other sources of pollution are not as obvious. Runoff from these same garbage disposal sites, seepage from waste treatment plants and illegal dumping all contribute to polluted lakes, rivers and groundwater. Precipitation causes agricultural waste, fertilizers and pesticides to run off farmland into water sources. Industrial gases such as carbon monoxide and sulfur dioxide from manufacturing and milling plants form *smog,* which in combination with precipitation forms *acid rain*. When this rain falls it contaminates not only our water resources but our land as well. It is evident in stunted tree growth, deteriorating masonry on buildings and clear lakes with no fish or lush vegetation. Industrial discharge from factories and production plants also contains such toxic elements as PCBs, dioxin, mercury, iron, lead, nickel, zinc, cadmium and arsenic. These contaminants pollute not only the location at which they are released but influence, in varying degrees, the ecosystem of the drainage basin all the way to the ocean.

What is a DEM (Digital Elevation Model)?

A DEM or Digital Elevation Model is a digital representation of elevation. Most imagery, at least in Remote Sensing applications, is taken from high altitudes looking straight down. It is often useful to simulate the view an observer would have if standing on the ground looking out over the image.

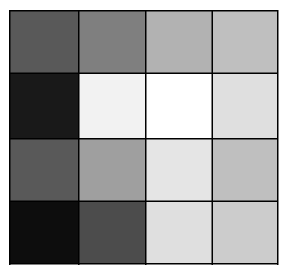
This is known as perspective scene. Those of you who play Nintendo or Play Station games have seen several of those perspective scenes. In order to create such a scene we need a DEM.

DEMs have many uses. Among the most important are the following:

- Storage of elevation data for digital topographic maps in national databases;
- Cut-and-fill problems in road design and other civil engineering projects;
- Three-dimensional display of land forms for military purposes (weapon guidance systems, pilot training) and for landscape design planning (landscape architecture);
- Planning routes for roads, locations of dams, etc.;
- Computing slope and aspect maps, and slope profiles that can be used to assist applications such predicting erosion hazards;
- As a background for displaying thematic information or for combining relief data with thematic data such as soils, land use or vegetation.

A Digital Elevation Model (DEM) as represented in the raster system, is simply an image layer where the values of each grid square represent the mean height, or elevation of land above some base plane. Sea level is commonly used as the base elevation. DEMs are commonly represented as the highest points being a bright colour or white, and the lowest points as dark colours or black. Each pixel will have a DEM value. The higher the value the higher is the elevation.

25	30	31	36
22	40	42	38
25	33	39	36
20	23	38	37



- ❖ **This information is taken from the Student Manual “Digital Image Processing Using EOScape” from PCI Enterprises**
- ❖ In the Teacher Tools Section of this lesson we are providing you with a Fly Demo of a DEM.